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## Apparatus and method for guiding an instrument by a luminous pointer


### Abstract

An instrument, a magnetic resonance imaging scanner and a method for operating the magnetic resonance imaging scanner with the instrument in which the instrument is aligned with a patient. The medical instrument comprises a first projector for projecting a luminous pointer in a predetermined alignment relative to the medical instrument. The magnetic resonance imaging scanner comprises a positioning aid configured to mark a predetermined position on an inner wall of a patient tunnel in a manner that is visible to a user. In the method, a predetermined alignment of the first projector relative to the medical instrument or position of the positioning aid on the wall of the patient tunnel is ascertained in which the luminous pointer of the first projector coincides with the positioning aid when the medical instrument is aligned parallel to a trajectory through an entry point on the patient and a target point in the patient and the medical instrument is aligned such that the luminous pointer and the positioning aid coincide.



**Inventors:** **Hengerer; Arne (Möhrendorf, DE)**

**Applicant:** **Siemens Healthcare GmbH** (Erlangen, DE)

**Family ID:** **80113177**

**Assignee:** **Siemens Healthineers AG (Forchheim, DE)**

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*Primary Examiner:* Buran; Ashley K

*Assistant Examiner:* Ly; Tommy T

*Attorney, Agent or Firm:* Lempia Summerfield Katz LLC

## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application claims the benefit of EP21217857.8, filed on Dec. 27, 2021, which is hereby incorporated by reference in its entirety.
FIELD
(2) Embodiments relate to an instrument, a magnetic resonance imaging scanner and a method for operating the magnetic resonance imaging scanner with the instrument in which the instrument is aligned with a patient.

## BACKGROUND

(3) Magnetic resonance imaging scanners are imaging apparatuses that, to map an object under examination, align nuclear spins of the object under examination with a strong external magnetic field and excite them to precess about this alignment by an alternating magnetic field. The precession or return of the spins from this excited state to a lower energy state in turn creates an alternating magnetic field in response which is received via antennas.

(4) Magnetic gradient fields are used to impart spatial encoding to the signals that subsequently provides the assignment of the received signal to a volume element. The received signal is then evaluated and a three-dimensional imaging representation of the object under examination is provided. The signal may be received using local receiving antennas, so-called local coils, that are arranged directly on the object under examination in order to achieve a better signal-to-noise ratio.

(5) A magnetic resonance imaging scanner provides visualization of the interior of the body over a longer period without exposing the patient or surgeon to an increased dose of ionizing radiation. However, instruments made of plastic or metal cannot be detected, or may only be detected indirectly, by magnetic resonance imaging and thus alignment is difficult.

## BRIEF SUMMARY AND DESCRIPTION

(6) The scope of the present invention is defined solely by the appended claims and is not affected to any degree by the statements within this summary. The present embodiments may obviate one or more of the drawbacks or limitations in the related art.

(7) Embodiments provide a method and an apparatus that provide for an instrument to be positioned and aligned with little additional effort.

(8) The medical instrument is intended for use on or in a patient. The instrument may, for example, be a biopsy needle, an instrument for ablation or for applying brachytherapy.

(9) The medical instrument includes a first projector for projecting a luminous pointer in a predetermined alignment relative to the medical instrument. A luminous pointer is considered to be a light pattern created by the projector on an inner surface of a patient tunnel that provides the projector, and thus the instrument, to be aligned with an angular error of less than 5 degrees, 2 degrees or 1 degree.

(10) According to the embodiments described below, the projector may, for example, be a simple light source with an optical system for alignment into a parallel beam, a laser with a collimation optical system, or also a projector that illuminates a pattern with a light source and projects it onto the inner wall. Variable projection by technology comparable to a beamer or by projection of an OLED matrix with an optical system may also be used.

(11) With the instrument, when the medical instrument is positioned on the patient appropriately for the application, the predetermined alignment of the first projector is directed away from the patient. In other words, when the medical instrument is arranged appropriately for the application, the luminous pointer of the projector does not strike the patient, but strikes the surrounding inner wall of the patient tunnel.

(12) The projector provides a second reference point for the alignment of the medical instrument to be created on the wall of the patient tunnel and used in addition to the contact point or entry point on the patient.

(13) The magnetic resonance imaging scanner includes a positioning aid. The positioning aid is configured to mark a predetermined position on an inner wall of a patient tunnel in a manner that is visible to a user. In other words, the positioning aid provides a user to identify a location on the inner wall of the tunnel, for example as a target for the light pointer of the first projector of the medical instrument. The positioning aid may, for example, be a single marking that differs in reflectivity from the surrounding inner wall. An elevation or depression on the surface of the inner wall may be used. However, temporary markings, which may be rendered visible variably by the magnetic resonance imaging scanner may be used.

(14) The positioning aids interact with the instrument to provide a defined alignment of the

instrument.
(15) The method relates to the positioning of an above-described medical instrument with a first projector for projecting a luminous pointer and a magnetic resonance imaging scanner with a positioning aid. The method is furthermore executed with a planning controller with an operator interface for planning an intervention with the medical instrument. Herein, the planning controller may be executed as a program on the magnetic resonance imaging scanner and use its output or also be executed on another computer or in a cloud as a separate unit.
(16) In one step of the method the patient is positioned in the magnetic resonance imaging scanner. This is done in such a way that a predetermined entry point for the medical instrument on the surface of the patient and a predetermined target point of the intervention in the patient are located in a detection area of the magnetic resonance imaging scanner. The detection area is also referred to as the field of view (FoV). Positioning may take place by the patient couch. The entry point may, for example, be marked with a magnetic resonance-active marker.
(17) In a further step of the method a predetermined alignment of the first projector relative to the medical instrument and/or position of the positioning aid on the wall of the patient tunnel is ascertained in which the luminous pointer of the first projector coincides with the positioning aid when the medical instrument is aligned parallel to a trajectory through the entry point and the target point. The idea is to achieve the alignment of the instrument in that a user aligns the luminous pointer with the positioning aid as explained in the following step. Either the position of the positioning aid to be targeted with the luminous pointer is changed in order to support or determine a changed alignment or the alignment of the first projector relative to the medical instrument is changed. Accordingly, in this step, the position of the positioning aid or the alignment of the first projector relative to the instrument under which the medical instrument is aligned parallel to the trajectory when the luminous pointer hits the positioning aid is determined. The calculation takes place in accordance with analytical geometry in the controller of the magnetic resonance imaging scanner, planning controller, or another computer by using rotations and determining a point of intersection of a straight line with a surface of the patient tunnel. The two parameters, position of the positioning aid, and alignment of the first projector, may be changed simultaneously and ascertained accordingly.
(18) In another step of the method the first projector is brought into the ascertained relative alignment with the medical instrument. This may, for example, take place by actuators on the projector that are connected to the controller of the magnetic resonance imaging scanner, for example via a wireless signal connection.
(19) Supplementarily or alternatively, the positioning aid is output. In the simplest case, this may entail outputting coordinates for the user on an output. Herein, the coordinates relate to a coordinate system that is attached in a visible manner to the inside of the patient tunnel and indicates a clear target position for the luminous pointer as a positioning aid for the user on the basis of the coordinates. However, a positioning aid that, for example, indicates a grid of light spots from which the light spot closest to the position of the luminous pointer to be achieved is activated may be used. For better compatibility with the magnetic resonance imaging scanner, the light spots may also be implemented by glass fibers with the light source being arranged away from the patient tunnel. A projection of the positioning aid would also be possible. The luminous pointer is activated if it is not already active.
(20) The method provides the user to be provided with a simple and reliable aid for the alignment of the medical instrument.
(21) In an embodiment of the medical instrument the medical instrument includes an axis of symmetry. The axis of symmetry may be a mirror-symmetrical axis or, for example, an axis for rotational symmetry.
(22) Aligning the first projector parallel to an axis of symmetry, for example an axis of rotation, provides the medical instrument to be held in different rotational positions without thereby

changing the alignment. In the case with rotationally symmetrical instruments, such as a biopsy needle, this simplifies alignment.

(23) In an embodiment of the medical instrument, the instrument includes a control input and is configured to change the predetermined alignment of the first projector in a predetermined manner in dependence on an instruction via the control input. Herein, alignment of the projector may be understood as the alignment of the projection of the luminous pointer with respect to the medical instrument. The first projector may be aligned mechanically relative to the instrument by actuators based on the instruction. However, the projector may also be configured to change the alignment of the luminous pointer, for example in that the projector relies on an image being projected with the luminous pointer and this image being changed in that the position of the luminous pointer is changed within the image. This may, for example, be achieved by projecting an OLED, DLP or LCD matrix. The control input may, for example, be an electrical or optical signal line or also a wireless data transmission, for example by Bluetooth or WLAN.

(24) If the alignment of the first projector is variable, one single positioning aid, or at least a small number of positioning aids, on the inner wall of the patient tunnel may be sufficient to ensure precise alignment.

(25) In an embodiment of the medical instrument, the instrument includes a sensor configured to detect an alignment of the instrument about the axis of symmetry relative to the magnetic resonance imaging scanner. For example, a sensor that ascertains the direction of gravity or of the B0 magnetic field.

(26) When the first projector is aligned relative to the instrument, the instrument may still be rotated by the user. If rotation about the axis of symmetry is determined by the sensor, the alignment of the first projector may be set in order to align the luminous pointer correctly taking account of the rotation.

(27) The luminous pointer and the positioning aid may be configured to identify, for example by a pattern and/or a shape of the luminous pointer, a rotation of the instrument about the axis of symmetry. In conjunction with a position marking, this may simultaneously indicate the alignment of the trajectory and rotation about the axis of symmetry and the user may thus set the alignment correctly. For example, the luminous pointer and positioning aid may include a directional arrow with a dot or circle at the origin. When the user causes the origin marking and the arrow to coincide, this provides the instrument to be aligned even if the axis of rotation of the instrument is different from the direction of projection of the first projector.

(28) In an embodiment of the magnetic resonance imaging scanner the positioning aid includes a marking attached in a visible manner to the inside of the patient tunnel. Positioning aids are markings that are different in color or reflection behavior or depressions or elevations that differ in shape from the surface. The positioning aid may also include a coordinate system on the inner wall of the patient tunnel.

(29) A visible positioning aid is advantageously simple to attach and use.

(30) In an embodiment of the magnetic resonance imaging scanner the magnetic resonance imaging scanner includes a second projector. The second projector is configured to mark a predetermined position on the inner wall of the patient tunnel in a visible manner by a light projection as a positioning aid. As described above with respect to the first projector, the second projector may be a laser or pointer that casts a single light spot onto the inner wall, or also a symbol or an entire pattern, such as a coordinate system. The second projector may be suitable for marking different positions, for example by projecting a pattern at different positions. This may, for example, be achieved in that the alignment of the second projector may be changed by the planning controller, or different images may be projected with the symbol or pattern at different positions.

(31) The second projector provides variable positions to be marked on the inner wall of the patient tunnel in order to achieve different alignments with a medical instrument.

(32) In an embodiment, the method further includes the step of aligning the medical instrument

such that the luminous pointer coincides with the positioning aid. The user of the medical instrument places the end or tip of the medical instrument that is opposite to the direction of projection of the first projector at the predetermined entry point. The entry point may be marked with an MR-active marker. The user then tilts and swivels the medical instrument such that the luminous pointer and the positioning aid are caused to coincide. If the luminous pointer and the positioning aid are not configured as rotationally symmetrical, but have a preferred direction to indicate rotation of the instrument about the trajectory, the luminous pointer and the positioning aid are aligned with one another as predetermined in order to achieve the desired orientation. For example, as described above, pointers may be caused to coincide.
(33) Alignment by the luminous pointer is a simple way of ensuring the desired alignment of the medical instrument with the target point.
(34) In an embodiment, the method furthermore includes the steps of providing a map of a patient on the operator interface of the planning controller and using the map to define an entry point on the surface of the patient and a target point in the patient by the user. The entry point in the map may be made visible by an MR active marker or made visible by deformation of the surface by the instrument or another object, such as, for example, a user's finger. The target point may be marked by the user in the map, for example made recognizable by a higher contrast due to higher density in a specific area. Automatic segmentation of the target point by the planning controller may also be possible.
(35) The trajectory for the method may be easily determined by selection on a map.
(36) In an embodiment of the method the alignment of the first projector with the medical instrument is fixed and is, for example, (anti-)parallel to a trajectory of the medical instrument.
(37) Then, in the step of ascertaining the alignment, only a position of the positioning aid on the wall of the patient tunnel is ascertained in which the luminous pointer of the first projector is caused to coincide with the positioning aid when the medical instrument is aligned parallel to a trajectory through the entry point and the target point. This is achieved in that a point of intersection of the luminous pointer with the inside of the is ascertained, for example by solving a system of equations with a straight line congruent to the trajectory and a mathematical description of the inner wall of the patient tunnel by a cylinder in the planning controller. The point of intersection is then output.
(38) The determination of the point of intersection is particularly simple if the trajectory is parallel to the direction of projection of the luminous pointer.
(39) In an embodiment of the method the point of intersection is output to the user as a coordinate at an operator interface.
(40) Herein, for example, a visible coordinate system is arranged on the inner wall of the patient tunnel. The coordinate system provides the user to clearly identify a position on the inner wall by the coordinates that are output to identify the point of intersection. The user may then target this position identified by the coordinates as a positioning aid with the illuminated viewfinder and thus achieve the desired alignment of the medical instrument.
(41) In an embodiment of the method the outputting of the point of intersection on the inside of the patient tunnel is a marking by the second projector. The second projector may be aligned by an actuator on instruction from the planning controller such that a projection of the second projector as a positioning aid marks the ascertained coordinates of the point of intersection on the inner wall of the patient tunnel. An image projected by the second projector onto the inner wall may be changed so that it marks the ascertained coordinate. It is also possible for a grid of light spots to be provided on the inner wall by LEDs or glass fibers, wherein the light spot closest to the ascertained point of intersection is activated by the planning controller as a positioning aid.
(42) Active marking of the variable point of intersection makes it easier for the user to find the target point for the luminous pointer and thus to align the instrument.
(43) In another embodiment of the method, it is possible to set the alignment of the first projector

relative to the instrument. Herein, in the step of ascertaining the alignment, an alignment of the first projector relative to the medical device is determined under which, on alignment of the medical instrument parallel to a trajectory through the entry point and the target point, the light pointer is aligned with the positioning aid. Since, herein, the alignment of the first projector is no longer parallel to the trajectory and thus dependent on the angle of rotation of the instrument about the trajectory. A sensor, such as, for example, a magnetic field sensor or a gravity sensor in the instrument may detect the angle and set the alignment of the first projector in dependence thereon.
(44) The first luminous pointer may not be rotationally symmetrical and may specify an angular position of the instrument if, as described above, the luminous pointer is aligned with an angular marking of the positioning aid. The alignment of the first projector relative to the instrument is then no longer dependent on a rotation of the instrument.
(45) Due to the degrees of freedom gained by a variable position of the first projector, the positioning aid may be provided in a clearly visible position, thus simplifying alignment for the user.
(46) The above-described properties, features and advantages, and the manner in which they are achieve will become clear and more plainly comprehensible in connection with the following description of embodiments, which will be explained in more detail in connection with the drawings.

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## Description

BRIEF DESCRIPTION OF THE FIGURES
(1) FIG. **1** depicts a schematic representation of a magnetic resonance imaging scanner with an instrument according to an embodiment.
(2) FIG. **2** depicts a schematic representation of an embodiment of a magnetic resonance imaging scanner with an instrument.
(3) FIG. **3** depicts a schematic representation of an embodiment of an instrument.
(4) FIG. **4** depicts a schematic representation of an embodiment of a magnetic resonance imaging scanner with an instrument.
(5) FIG. **5** depicts a schematic representation of an embodiment of an instrument.
(6) FIG. **6** depicts a schematic representation of an embodiment of a magnetic resonance imaging scanner.
(7) FIG. **7** depicts a schematic representation of an embodiment of an instrument.
(8) FIG. **8** depicts a schematic flow diagram of a method according to an embodiment.
DETAILED DESCRIPTION
(9) FIG. **1** depicts a schematic representation of an embodiment of a magnetic resonance imaging scanner **1** for executing the method.
(10) The magnet unit **10** includes a field magnet **11** that creates a static magnetic field B0 for aligning nuclear spins of samples or the patient **100** in a recording area. The recording area is characterized by an extremely homogeneous static magnetic field B0. The homogeneity relates to the magnetic field strength or the magnitude.
(11) The recording area is almost spherical and arranged in a patient tunnel **16** that extends in a longitudinal direction **2** through the magnet unit **10**. A patient couch **30** may be moved in the patient tunnel **16** by the drive unit **36**. The field magnet **11** may be a superconducting magnet able to provide magnetic fields with a magnetic flux density of up to 3T, or even more in the case of the latest devices. However, for lower magnetic field strengths it is also possible to use permanent magnets or electromagnets with normally conducting coils.
(12) The magnet unit **10** furthermore includes gradient coils **12** configured to superimpose temporally and spatially variable magnetic fields in three spatial directions on the magnetic field

B0 for spatial differentiation of the acquired mapping regions in the examination volume. The gradient coils **12** may be coils made of normally conducting wires that may create mutually orthogonal fields in the examination volume.

(13) The magnet unit **10** likewise includes a body coil **14** configured to radiate a radio-frequency signal supplied via a signal line **33** into the examination volume and to receive resonance signals emitted by the patient **100** and output them via a signal line **33**.

(14) A control unit **20** supplies the magnet unit **10** with the various signals for the gradient coils **12** and the body coil **14** and evaluates the received signals.

(15) Thus, the control unit **20** includes a gradient actuator **21** configured to supply the gradient coils **12** with variable currents via supply lines which provide the desired gradient fields in the examination volume in a time-coordinated manner.

(16) The control unit **20** includes a radio-frequency unit **22** configured to create a radio-frequency pulse with a prespecified temporal course, amplitude and spectral power distribution for exciting a magnetic resonance of the nuclear spins in the patient **100**. Herein, pulse powers in the kilowatt range may be achieved. The excitation signals may be radiated into the patient **100** via the body coil **14** or also via a local transmitting antenna **50**.

(17) A controller **23** communicates with the gradient controller **21** and the radio-frequency unit **22** via a signal bus **25**.

(18) A medical instrument **60**, for example a biopsy needle, is arranged on a patient **100** in a patient tunnel **16**.

(19) FIG. **2** shows a detail of an embodiment in FIG. **1**.

(20) There is a predetermined target point **103** in the patient, for example a node, that was identified with its coordinates in a map of the patient **100**. A medical instrument **60**, for example a biopsy needle, with which a sample is to be taken from the target object at the target point **103** is placed on an entry point on the surface of the patient which is indicated by a marking **102**. The marking may also be visible in a magnetic resonance image. However, a marking **102** formed solely by the impression made by the instrument **60** in the patient may also be visible in the magnetic resonance image. A trajectory **101**, a straight line, connects the entry point and the target point **103**. The trajectory may be selected such that it does not hit any important organ, blood vessel, nerve, or other tissue to be protected.

(21) The predetermined trajectory **101** includes a point of intersection with the patient tunnel **16** or the inner wall thereof on a side facing away from the patient **100**. If the relative position of the patient **100** to the magnetic resonance imaging scanner **1** is known, this point of intersection may be calculated using the methods of linear algebra. One possibility for determining the trajectory **101** and point of intersection is provided by a magnetic resonance recording of the patient **100** in which the target object **103** and the planned entry point is visible. The point of intersection calculated by the controller **23**, for example, may be output to a user in the form of coordinates for a coordinate system **71** attached to the inner wall in a visible manner. This provides the user to identify the predetermined position **80** on the inner wall of the patient tunnel **16** where the trajectory **101** intersects the inner wall. The user may align the instrument **60** in FIG. **2** parallel to the trajectory **101** and thus to the target object **103** if, as shown in more detail in FIG. **3**, this projects a light spot or a luminous pointer **70** and the user causes this luminous pointer **70** to coincide with the predetermined position **80**. The coordinate system **71** may be projected onto the inner wall of the patient tunnel **16** by a second projector **90**.

(22) FIG. **3** is a schematic representation of an embodiment of the medical instrument. The medical instrument **60** includes an axis of symmetry, defined here by the biopsy needle **61**. The symmetry to the axis of symmetry reflects a symmetry of the medical instrument **60** or the application to be performed therewith. In the case of the biopsy needle, this may be rotated about an axis through the longitudinal extent of the hollow needle **61** during the application without further consequences. Accordingly, the axis of symmetry of the medical instrument **60** in FIG. **2** is an axis of rotation

through the longitudinal extent of the hollow needle **61**. If this is caused to coincide with the predetermined trajectory **101**, the alignment of the medical instrument **60** is aligned correctly with the target object **103**.
(23) For this purpose, the medical instrument **60** in FIG. **3** includes a first projector **72**. The first projector **72** includes a power source, for example a battery **73**. However, the power source may also be a power supply line. The power source supplies a light source, for example, a semiconductor laser **74**. However, other light sources, such as lamps or an OLED matrix may be used. The light created is focused by an optical system **75** into a light beam that is emitted parallel to the axis of symmetry of the medical instrument **60** in a direction opposite to the patient **100**. When the light beam strikes a surface, for example the inner wall of the patient tunnel **16**, it creates a light pointer **70** there in the form of a dot or spot or a more complex pattern, as detailed below.
(24) If the medical instrument **60** is arranged with the tip or the hollow needle **61** at the entry point and the user aligns the luminous pointer **70** with the predetermined position **80**, these two fixed points define a straight line, which corresponds to the predetermined trajectory **101** and ensures alignment of the medical instrument **60** in accordance with the invention.
(25) FIG. **4** is a representation of an embodiment of the medical instrument **60** and the magnetic resonance imaging scanner **1**. The same reference numbers denote the same objects, for example the trajectory **101** is still defined as the straight line through the entry point and target object **103**. The embodiment represented in FIG. **4** essentially differs in that the trajectory **101** no longer matches the direction of projection of the luminous pointer **70**. This makes it possible to no longer select constantly changing coordinates on the inside of the patient tunnel **16** as the predetermined position **80**, but only to mark one or a few predetermined positions **80**, which are then, for example, marked by a fixed marking, such as an elevation or depression or reflex point. However, the alignment of the first projector **72** relative to the medical device **60** has to be changed in each case, as represented in detail in FIG. **5**.
(26) In FIG. **5**, the first projector **72** is no longer aligned parallel to the axis of symmetry of the medical instrument **60**, but may be changed by a first actuator **77** and an optional second actuator **78** relative to the axis of symmetry. For example, a projector controller **76** in the medical instrument may receive setting instructions, for example wirelessly, from the controller **23** of the magnetic resonance imaging scanner **1**. The control instructions may specify an angular alignment with respect to the axis of symmetry by the first actuator **77** and/or an angular alignment with respect to a rotation about the axis of symmetry by the second actuator **78**. The first actuator and the second actuator may, for example, be motors, for example piezoelectric actuators, that execute a linear movement or a rotation without requiring a disruptive magnetic field.
(27) Herein, the angles to be set depend upon a rotation of the instrument **60** about the axis of symmetry relative to the magnetic resonance imaging scanner or also the direction of gravity. This rotation may be detected by the projector controller **76** by a sensor **79** and transmitted to the controller **23** for ascertaining the angles for the first actuator **77** and the second actuator **78**. The sensor **79** may, for example, be a MEMS-based gravity sensor or a multi-dimensional Hall sensor.
(28) The first projector **72** may be configured as freely rotatable about the axis of rotation and the weight distribution may be set such that the first projector **72** has a preferred direction under the effect of gravity, for example directed upward against the force of gravity.
(29) FIG. **6** is a representation of two possibilities for marking the predetermined position **80** in a magnetic resonance imaging scanner **1**.
(30) A second projector **90** is attached in or on the patient tunnel **16** and projects into the interior thereof. The second projector **90** may, for example, project an image with a variable position and/or alignment as a predetermined position **80** for the luminous pointer **70**. The image may, for example, be created with an OLED, LCD or DLP matrix. The second projector **90** may be variably aligned using actuators, as was explained in FIG. **5** with respect to the first projector **72**.
(31) However, a variable marking of the predetermined position **80** may also be created, for

example by a switchable matrix of light spots **91** on the inner wall of the patient tunnel **16**, which are, for example, created by LEDs and optionally transmitted by light guides to the inner wall of the patient tunnel **16**.
(32) FIG. **7** is a schematic representation of an embodiment of a medical instrument **60**. The embodiment represented represents an alternative to the variant represented in FIG. **5**. Instead of aligning the first projector **72** differently by a first actuator **77** and/or second actuator **78**, the projected image is changed. The first projector **72** includes a projection angle that is, for example, greater than 20 degrees, 40 degrees or 60 degrees. This may cause the luminous pointer **70** to coincide with the predetermined position **80** in many different alignments of the trajectory **101**. The luminous pointer **70** marks a position and a direction, for example by the point of origin and the alignment of the arrow in FIG. **7**. Causing this luminous pointer **70** to coincide with a predetermined position **80**, likewise marked by a location marking and a direction marking, may also achieve a defined position of the medical instrument **60** for rotation about the axis of symmetry.
(33) FIG. **8** depicts a schematic flow diagram of a method.
(34) In a step S**30**, the patient **100** is positioned in the magnetic resonance imaging scanner **1**, for example with the patient table **30**. The patient **100** is to undergo a minimally invasive examination or an intervention by a medical instrument **60** on an organ or tissue, that is referred to below as the target object **103**. The way in which this entry point may be determined is explained below. Herein, the positioning takes place such that the entry point and the target object **103** are located in a detection area of the magnetic resonance imaging scanner **1**, that is also referred to as the field of view (FoV). The correct position may, for example, be verified using markings **102** on the patient **100** and magnetic resonance imaging scanner **1** or fast image acquisition using the magnetic resonance imaging scanner **1**. At the end of the positioning, the patient **100** is in a known relative position and alignment relative to the magnetic resonance imaging scanner **1**.
(35) In another step S**40** of the method a predetermined alignment of the first projector **72** relative to the medical instrument **60** and/or position of the positioning aid on the wall of the patient tunnel **16** is ascertained in which the luminous pointer **70** of the first projector **72** coincides with the positioning aid when the medical instrument is aligned parallel to a trajectory **101** through the entry point and the target point.
(36) Step S**40** differs for cases in which the first projector **72** is aligned parallel to the axis of symmetry of the medical instrument, as in FIG. **3**, or may itself be aligned with the axis of symmetry, as in FIG. **5** or FIG. **7**.
(37) For the embodiment in FIG. **3** with an axis of symmetry parallel to the trajectory **101**, a point of intersection of the trajectory through the predetermined entry point and target object **103** with the inner wall of the patient tunnel **16** on a side of the instrument **60** facing away from the patient **100** is determined in a step S**41**. This may, for example, take place by solving a system of equations in which the straight-line equation of the trajectory **101** is equated with an envelope of the inner wall.
(38) In the embodiments in FIGS. **5** and **7**, an alignment of the first projector **72** may be set at an angle relative to the axis of symmetry of the medical instrument **60** and an angle of rotation about the axis of symmetry. The predetermined position **80** may be arranged in an invariable manner on the inner wall of the patient tunnel **16** and marked graphically or by a structure. In this case, the alignment of the first projector **72** is determined in a step S**42** such that the luminous pointer **70** of the first projector falls onto the predetermined position **80** when the medical instrument **60** is aligned parallel to the trajectory **101** in a manner appropriate for the application. This may be done by solving a system of equations in which the trajectory **101** is multiplied by a rotation matrix containing as variables the two angles of alignment of the first projector **72** and that is equated with the predetermined position **80**. The solution according to the two angles provides the alignment of the first projector **72** to be determined, for example by the controller **23** of the magnetic resonance

imaging scanner **1**. during the ascertaining, the projector controller **76** may ascertain, by a sensor **79**, a position of the medical instrument **60** relative to the magnetic resonance imaging scanner **1** and the alignment is ascertained in dependence on this position. For example, this position may be included as an offset in the angle of rotation.

(39) In a step S**50**, the result of the ascertaining in step S**40** is output.

(40) For an embodiment of the medical instrument **60** with fixed parallel alignment of the first projector **72** and the luminous pointer **70** created thereby with the axis of symmetry, the coordinate for the predetermined position **80** may be output in a step S**51** on an output of the magnetic resonance imaging scanner **1**. A second projector **90** may indicate the predetermined position **90** in that the second projector **90** is aligned by actuators with the predetermined position **80** by the controller **23**. The second projector **90** may be aligned in a fixed manner, but outputs a variable image with a settable position of a projected marking for the predetermined position **80** which is, for example, set by the controller **23**.

(41) For an embodiment in which the first projector **72**, or at least the luminous pointer **80** projected thereby, is variable in alignment relative to the medical instrument **60**, the outputting takes place in a step S**52**, for example in that the first actuator **77** and second actuator **78** are aligned accordingly by a setting instruction from the controller **23** to the projector controller **76**.

(42) In an embodiment with a fixed first projector **72** that projects an image, the projector controller **76** sets the image such that the luminous pointer **70** is projected in the ascertained alignment relative to the medical instrument **60**.

(43) The luminous pointer **70** is then projected by the first projector **72**.

(44) In another step S**60**, the medical instrument **60** is positioned at the predetermined entry point. This may be done by a user and may also be done earlier in the method.

(45) Finally, in a step S**60**, the medical instrument is aligned such that the luminous pointer **70** is caused to coincide with the positioning aid. This may be done by the user by tilting, swiveling and/or rotating the medical instrument **60** until the light pointer **70** coincides with the predetermined position **80**. The tip of the medical instrument **60** remains at the entry point. In an embodiment in which the light pointer **70** and the marking for the predetermined position **80** includes a direction marking for displaying directional information, the alignment of the medical instrument **60** may also include rotating the medical instrument **60** about the axis of symmetry until the direction markings coincide.

(46) In an embodiment, the method furthermore includes the step S**10** of outputting a map of the patient **100** on the operator interface of the planning controller. The planning controller may be identical to the controller **23** of the magnetic resonance imaging scanner **1**. The map contains at least the target object **103** and an area around a possible entry point. The map may be a magnetic resonance scan with the magnetic resonance imaging scanner **1**. The map may be acquired with another modality and may be provided on a data carrier or via a data network connection.

(47) In a further step S**20**, an entry point is defined on the surface of the patient and a target point is defined in the target area **103** in the patient **100**. The target point may be defined by an organ or tissue identifiable in the map, for example a node or an entire organ. The entry point may already be marked in the map by a marker detected by the modality. The entry point may only be defined based on the map, for example with the requirement that no sensitive tissue is damaged on the way from the entry point to the target area.

(48) It is to be understood that the elements and features recited in the appended claims may be combined in different ways to produce new claims that likewise fall within the scope of the present invention. Thus, whereas the dependent claims appended below depend from only a single independent or dependent claim, it is to be understood that these dependent claims may, alternatively, be made to depend in the alternative from any preceding or following claim, whether independent or dependent, and that such new combinations are to be understood as forming a part of the present specification.

(49) While the present invention has been described above by reference to various embodiments, it may be understood that many changes and modifications may be made to the described embodiments. It is therefore intended that the foregoing description be regarded as illustrative rather than limiting, and that it be understood that all equivalents and/or combinations of embodiments are intended to be included in this description.

## Claims

1. A method for positioning a medical instrument on a patient with a magnetic resonance imaging scanner that includes a positioning aid configured to mark a predetermined position on an inner wall of a patient tunnel in a manner that is visible to a user, the medical instrument comprising a first projector configured for projecting a luminous pointer in a predetermined alignment relative to the medical instrument, the method comprising: positioning the patient in the magnetic resonance imaging scanner such that a predetermined entry point for the medical instrument on a surface of the patient and a predetermined target point of an intervention in the patient are located in a detection area of the magnetic resonance imaging scanner; ascertaining a predetermined alignment of the first projector relative to the medical instrument in which the luminous pointer of the first projector coincides with the positioning aid when the medical instrument is aligned parallel to a trajectory through the entry point and the target point or ascertaining a position of the positioning aid on the wall of the patient tunnel, in which the luminous pointer of the first projector coincides with the positioning aid when the medical instrument is aligned parallel to the trajectory through the entry point and the target point; aligning the first projector or outputting coordinates that relate to a coordinate system that is attached to the inside wall of the patient tunnel in a visible manner to the user; and projecting the luminous pointer.
2. The method of claim 1, wherein the method further comprises: aligning the medical instrument such that the luminous pointer coincides with the positioning aid.
3. The method of claim 1, wherein the method further comprises: providing a map of the patient on an operator interface of a planning controller; and defining an entry point on the surface of the patient and a target point in the patient.
4. The method of claim 1, wherein an alignment of the first projector with the medical instrument is fixed and parallel to an axis of symmetry of the medical instrument, and wherein a point of intersection of the luminous pointer is determined and the point of intersection is output.
5. The method of claim 1, wherein an alignment of the first projector with the medical instrument is set by the magnetic resonance imaging scanner, and for ascertaining the alignment, an alignment of the first projector relative to the medical instrument is determined under which, on alignment of the medical instrument parallel to a trajectory through the entry point and the target point, the luminous pointer is aligned with the positioning aid, and the first projector is aligned according to the determined alignment.